Analysis of Car Price and Economy in Qatar

STAT231: Applied Regression Analysis Course Project

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Overview

This report presents an analysis of the fuel economy and price of cars in Qatar using the `qatarcars` dataset that was developed by Musgrave (2025). We develop multiple linear regression models to understand the relationships between car specifications and their fuel economy and price. The analysis includes model diagnostics, visualizations, and performance evaluation. The key findings indicate that horsepower, mass, and engine type are significant predictors of fuel economy, while horsepower and performance are significant predictors of price. The models demonstrate reasonable predictive performance.

1 Introduction

In this project, we analyze the fuel economy and price of cars in Qatar using the `qatarcars` dataset. We develop multiple linear regression models to understand the relationships between car specifications and their fuel economy and price. The analysis includes model diagnostics, visualizations, and performance evaluation.

2 Dataset description

The **qatarcars** dataset includes fuel economy, size, performance, and price data for cars in Qatar. Data was sourced from YallaMotors Qatar between January and August 2025.

Key Variables:

- `economy`: Fuel economy (L/100km)
- `price`: Car price (2025 Qatari Riyals)
- `horsepower`: Engine power
- `mass`: Car mass (kg)
- `performance`: 0-100 km/h acceleration time (seconds)
- `origin`: Country of manufacture
- `enginetype`: Type of engine

The dataset contains 94 observations after removing missing values.

3 Setup

```r
set.seed(123)
library(tidyverse)
library(car)
library(knitr)
library(qatarcars)
library(caret)
library(randtests)
library(patchwork)

# Color palette
colors <- c("#E63946", "#2196F3", "#FF9800", "#4CAF50", "#9C27B0",
            "#00BCD4", "#FF5722", "#8BC34A", "#673AB7", "#F06292")
```

4 Data Cleaning and Exploration

4.1 Initial Data Exploration

```r
data <- qatarcars
attach(data)

# Checking structure and summary of the dataset
glimpse(data)
```

```
Rows: 105
Columns: 15
$ origin      <fct> Germany, Germany, Germany, Germany, Germany, Germany, Germ~
$ make        <fct> BMW, BMW, Audi, Audi, Audi, Mercedes, Mercedes, Mercedes, ~
$ model       <fct> 3 Series Sedan, X1, RS Q8, RS3, A3, Maybach, G-Wagon, EQS,~
$ length      <dbl> 4.713, 4.505, 5.012, 4.542, 4.456, 5.469, 4.613, 5.216, 4.~
$ width       <dbl> 1.827, 1.845, 1.694, 1.851, 1.960, 1.921, 1.984, 1.926, 1.~
$ height      <dbl> 1.440, 1.642, 1.998, 1.412, 1.416, 1.510, 1.969, 1.512, 1.~
$ seating     <dbl> 5, 5, 5, 5, 5, 4, 5, 5, 5, 5, 4, 5, 5, 5, 5, 5, 5, 2, 2, 5~
$ trunk       <dbl> 59, 505, 605, 321, 425, 500, 480, 610, 435, 565, 132, 627,~
$ economy     <dbl> 7.6, 6.6, 12.1, 8.7, 6.5, 13.3, 13.1, NA, 5.6, 7.5, 11.2, ~
$ horsepower  <dbl> 386, 313, 600, 400, 180, 612, 585, 333, 163, 221, 473, 453~
$ price       <dbl> 164257, 264000, 630000, 310000, 165000, 1281000, 1011500, ~
$ mass        <dbl> 1653, 1701, 2490, 1565, 1325, 2376, 2588, 2495, 1565, 1656~
$ performance <dbl> 4.3, 5.4, 3.6, 3.8, 6.7, 4.1, 4.3, 5.6, 6.8, 9.0, 3.3, 4.4~
$ type        <fct> Sedan, SUV, SUV, Sedan, Sedan, Sedan, SUV, SUV, SUV, SUV, ~
$ enginetype  <fct> Petrol, Petrol, Petrol, Petrol, Petrol, Petrol, Petrol, El~
```

```r
colnames(data)
```

```
 [1] "origin"      "make"        "model"       "length"      "width"
 [6] "height"      "seating"     "trunk"       "economy"     "horsepower"
[11] "price"       "mass"        "performance" "type"        "enginetype"
```

4.2 Missing values summary

```r
# NA summary
sub <- sapply(data, function(x) sum(is.na(x)))

kable(as.data.frame(sub) |> filter(sub > 0), col.names = c("Variable", "NA_Count"))
```

Table 1: Missing Values Summary

Variable	NA_Count
economy	10

```r
# Missing values in economy variable detected
# economy has 10 NAs
data |> filter(is.na(economy))
```

```
# A tibble: 10 x 15
   origin      make   model length width height seating trunk economy horsepower
   <fct>       <fct>  <fct>  <dbl> <dbl>  <dbl>   <dbl> <dbl>   <dbl>      <dbl>
 1 Germany     Merce~ EQS     5.22  1.93   1.51       5   610      NA        333
 2 South Korea Hyund~ Ioni~   4.64  1.89   1.60       5   527      NA        302
 3 South Korea Kia    EV9     5.01  1.98   1.75       7   572      NA        215
 4 UK          Jaguar I-Pa~   4.68  2.01   1.56       5   656      NA        395
 5 UK          Lotus  Evija   4.46  2      1.12       2     0      NA       1973
 6 USA         Tesla  Mode~   5.06  2.27   1.68       5  1233      NA        670
 7 USA         Tesla  Mode~   4.75  1.92   1.62       5   480      NA        342
 8 USA         Tesla  Mode~   4.72  1.85   1.44       5   542      NA        460
 9 PR China    BYD    Seal    4.8   1.88   1.46       5   400      NA        204
10 PR China    BYD    Atto~   4.46  1.88   1.62       5   440      NA        201
# i 5 more variables: price <dbl>, mass <dbl>, performance <dbl>, type <fct>,
#   enginetype <fct>
```

```r
# isna vs enginetype
table(is.na(economy), enginetype)
```

```
      enginetype
       Electric Hybrid Petrol
  FALSE        0     14     81
  TRUE        10      0      0
```

> **Note**
>
> - `economy` variable has 10 missing values, which are all for electric cars.
> - This is a case of structural missingness, as fuel economy in L/100km is not applicable for electric vehicles.
> - We will remove these observations from the analysis for economy but keep them for price analysis.

4.3 Summary statistics

```r
# summary of continuous variables
#| tbl-cap: "Summary Statistics for Economy, Price, Horsepower, and Mass"
kable(summary(data |> dplyr::select(economy, price, horsepower, mass)))
```

economy	price	horsepower	mass
Min. : 4.100	Min. : 35000	Min. : 76.0	Min. : 945
1st Qu.: 6.500	1st Qu.: 91000	1st Qu.: 154.0	1st Qu.:1428
Median : 7.600	Median : 164000	Median : 248.0	Median :1701
Mean : 8.743	Mean : 796573	Mean : 317.8	Mean :1776
3rd Qu.:10.650	3rd Qu.: 310000	3rd Qu.: 380.0	3rd Qu.:2055
Max. :22.500	Max. :33000000	Max. :1973.0	Max. :2746
NA's :10	NA	NA	NA

> **Note**
>
> - extreme skewness observed for price variable

```r
kable(summary(data |> dplyr::select(length, width, height, trunk, seating, performance)))
```

Table 2: Summary Statistics for Dimensions and Performance Variables

length	width	height	trunk	seating	performance
Min. :3.595	Min. :1.595	Min. :1.122	Min. : 0.0	Min. :2.00	Min. : 2.400
1st Qu.:4.495	1st Qu.:1.818	1st Qu.:1.455	1st Qu.: 284.0	1st Qu.:5.00	1st Qu.: 4.800
Median :4.681	Median :1.875	Median :1.540	Median : 448.0	Median :5.00	Median : 6.900

length	width	height	trunk	seating	performance
Mean :4.664	Mean :1.887	Mean :1.574	Mean : 437.6	Mean :5.01	Mean : 7.101
3rd Qu.:4.850	3rd Qu.:1.976	3rd Qu.:1.685	3rd Qu.: 542.0	3rd Qu.:5.00	3rd Qu.: 8.800
Max. :5.469	Max. :2.271	Max. :1.998	Max. :1233.0	Max. :8.00	Max. :14.500

```r
# summary of categorical variables
```

```r
kable(table(data$origin, data$enginetype))
```

Table 3: Cross-tabulation of Origin and Engine Type

	Electric	Hybrid	Petrol
Germany	1	1	18
Italy	0	0	5
Japan	0	5	24
PR China	2	0	16
South Korea	2	1	7
Sweden	0	3	0
UK	2	4	5
USA	3	0	6

```r
kable(table(data$origin, data$type))
```

Table 4: Cross-tabulation of Origin and Car Type

	Coupe	Hatchback	SUV	Sedan
Germany	3	1	12	4
Italy	2	0	1	2
Japan	4	2	14	9
PR China	0	2	10	6
South Korea	0	2	4	4
Sweden	0	0	3	0
UK	5	0	5	1
USA	0	1	6	2

```r
data_with_electric <- data |>
    mutate(origin2 = case_when(
        origin %in% c("Germany", "Italy", "Sweden", "UK") ~ "European",
        origin %in% c("Japan", "PR China", "South Korea", "USA") ~ "Non-European"
    ))

data<- data |>
    filter(!is.na(economy)) |>
```

```r
    mutate(origin2 = case_when(
        origin %in% c("Germany", "Italy", "Sweden", "UK") ~ "European",
        origin %in% c("Japan", "PR China", "South Korea", "USA") ~ "Non-European"
    ))


# table with model and make as columns and count as values
```

> **Note**
>
> - Extracted `origin2` variable for better analysis, categorizing cars into European and Non-European based on their country of manufacture.
> - This categorization allows us to analyze differences in fuel economy and price based on the broader origin of the cars, which may capture regional differences in design and engineering.

```r
kable(data |> group_by(make) |>
    summarise(count = n(),
              price = round(mean(price), 2),
              economy = round(mean(economy), 2)))
```

Table 5: Summary Statistics by Make

make	count	price	economy
Audi	3	368333.33	9.10
Bentley	3	1146666.67	11.07
Bestune	2	47000.00	7.15
BMW	5	278551.40	6.92
Bugatti	2	23051980.00	22.40
Cadillac	3	219200.00	11.53
Chery	3	85400.00	6.73
Ford	3	185133.33	10.40
Geely	3	110800.00	6.80
Honda	3	123999.00	7.90
Hyundai	3	65566.67	6.83
Jetour	3	101333.33	8.97
Kia	5	105800.00	7.10
Land Rover	3	219700.00	9.87
Lexus	4	331750.00	10.95
Maserati	3	369833.33	8.90
Mazda	4	99750.00	6.68
McLaren	3	2355233.33	9.23
Mercedes	4	667749.25	9.88
MG	5	67600.00	8.00
Mitsubishi	4	67750.00	6.00
Nissan	3	148500.00	11.20
Porsche	3	346166.67	13.73

make	count	price	economy
Subaru	1	125000.00	9.50
Toyota	10	128630.00	6.92
Volkswagen	4	131700.00	7.50
Volvo	3	198333.33	7.30

Summary of Data Exploration

- The dataset has `NA` for the `economy` variable, which is only present for electric cars. We will remove it from the analysis for economy but keep it for price analysis.
- The 'price' variable is heavily right-skewed due to luxury cars, so we can consider log transformation for price in the analysis.
- The 'origin' variable can be categorized into three groups: European, USA, and Asian for better analysis. All the brands come under this categorization.
- There are some outliers in economy as well, which will be handled during the analysis.

5 Economy Analysis

5.1 Model Development

```
# box plot for economy based on origin2
ggplot(data, aes(x = origin2, y = economy, fill = origin2)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values = colors[c(1, 2, 3)]) +
  labs(title = "Fuel Economy by Origin", x = "Origin", y = "L/100km") +
  theme_minimal() +
  theme(legend.position = "none")
```

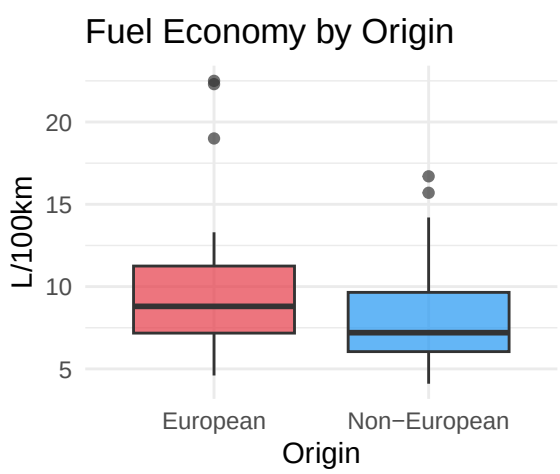


Figure 1: Fuel Economy by Origin - Box Plot

```
# all continuous variables + selected categorical variable
# origin, make and model are having too many levels
# type, enginetype, origin2
fit1 <- lm(economy ~ length + width + height + seating + trunk +
    horsepower + price + mass + performance + origin2 +
    type + enginetype, data = data)
summary(fit1)
```

```
Call:
lm(formula = economy ~ length + width + height + seating + trunk +
    horsepower + price + mass + performance + origin2 + type +
    enginetype, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
```

```
-4.2720 -1.1242 -0.0571  0.8163  8.2802

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)         -1.226e+01  6.223e+00  -1.970 0.052317 .
length              -1.353e-01  1.147e+00  -0.118 0.906387
width                4.861e+00  2.751e+00   1.767 0.081100 .
height               1.916e+00  2.954e+00   0.649 0.518384
seating             -4.917e-01  3.098e-01  -1.587 0.116464
trunk               -1.701e-03  9.463e-04  -1.798 0.075957 .
horsepower           7.084e-03  2.224e-03   3.185 0.002064 **
price                1.137e-07  1.041e-07   1.092 0.278108
mass                 3.612e-03  8.853e-04   4.080 0.000106 ***
performance          1.009e-01  1.296e-01   0.778 0.438742
origin2Non-European  1.280e+00  5.477e-01   2.338 0.021898 *
typeHatchback        9.909e-01  1.263e+00   0.785 0.434937
typeSUV              6.756e-01  1.304e+00   0.518 0.605797
typeSedan            1.251e+00  1.003e+00   1.248 0.215785
enginetypePetrol     2.213e+00  5.824e-01   3.800 0.000281 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.886 on 80 degrees of freedom
Multiple R-squared:  0.7548,    Adjusted R-squared:  0.7119
F-statistic: 17.59 on 14 and 80 DF,  p-value: < 2.2e-16
```

```r
# stepwise selection
fit2 <- step(fit1, direction = "both")
# not printed ~ long summary
```

```r
summary(fit2)
```

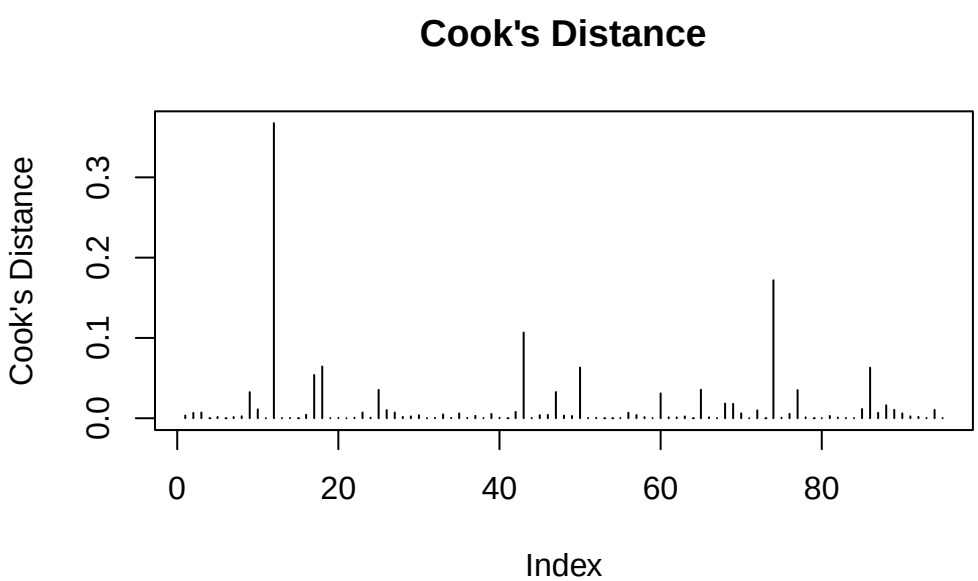
```
Call:
lm(formula = economy ~ horsepower + mass + origin2 + enginetype,
    data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-5.7289 -0.9411 -0.0379  0.9756  8.4153

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)         -2.1291560  1.0812396  -1.969  0.05201 .
horsepower           0.0094058  0.0009180  10.246  < 2e-16 ***
mass                 0.0032312  0.0004615   7.001 4.42e-10 ***
origin2Non-European  1.1915262  0.4610869   2.584  0.01137 *
enginetypePetrol     2.0072233  0.5592720   3.589  0.00054 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.876 on 90 degrees of freedom
Multiple R-squared:  0.7268,    Adjusted R-squared:  0.7147
F-statistic: 59.86 on 4 and 90 DF,  p-value: < 2.2e-16
```

```r
shapiro.test(fit2$residuals)
```

```
    Shapiro-Wilk normality test

data:  fit2$residuals
W = 0.93535, p-value = 0.000153
```

```r
# not normal, checking for outliers


# not normal, checking for outliers
cooksd <- cooks.distance(fit2)
plot(cooksd, type = "h", main = "Cook's Distance", ylab = "Cook's Distance")
```




```r
which(cooksd > 4/nrow(data))
```

```
12 17 18 43 50 74 86
12 17 18 43 50 74 86
```

```r
outs <- c(17, 18, 74)
data[outs, ]
```

```
# A tibble: 3 x 16
  origin make  model length width height seating trunk economy horsepower  price
  <fct>  <fct> <fct>  <dbl> <dbl>  <dbl>   <dbl> <dbl>   <dbl>      <dbl>  <dbl>
1 Italy  Buga~ Chir~   4.54  2.04   1.21       2    45    22.3       1578 1.31e7
2 Italy  Buga~ Cent~   4.54  2.05   1.2        2     0    22.5       1600 3.30e7
3 UK     McLa~ Artu~   4.54  1.98   1.19       2     0     4.6        671 9.46e5
# i 5 more variables: mass <dbl>, performance <dbl>, type <fct>,
#   enginetype <fct>, origin2 <chr>
```

```r
data |> filter(price > 1e6)
```

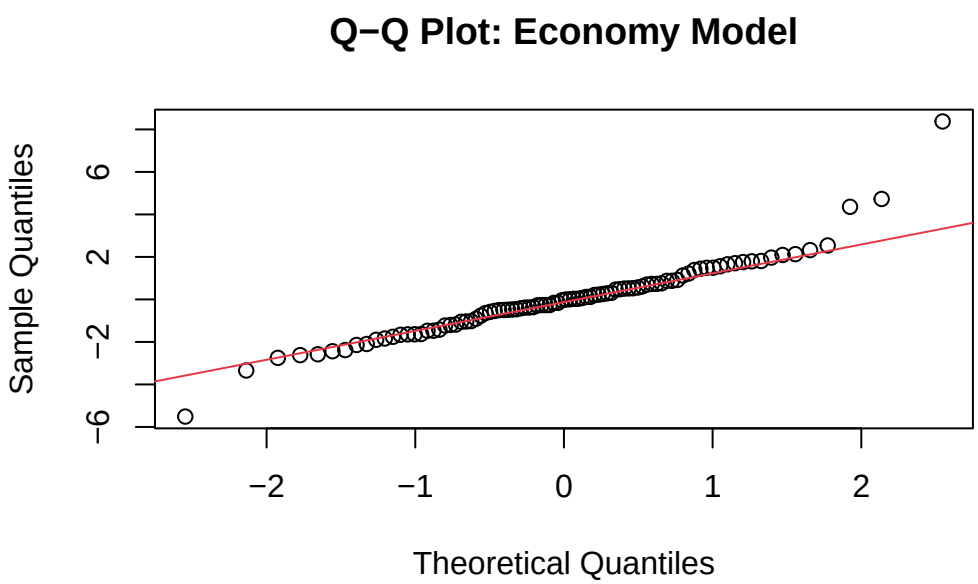
```
# A tibble: 8 x 16
  origin make  model length width height seating trunk economy horsepower  price
  <fct>  <fct> <fct>  <dbl> <dbl>  <dbl>   <dbl> <dbl>   <dbl>      <dbl>  <dbl>
1 Germa~ Merc~ Mayb~   5.47  1.92   1.51       4   500    13.3        612 1.28e6
2 Germa~ Merc~ G-Wa~   4.61  1.98   1.97       5   480    13.1        585 1.01e6
3 Italy  Buga~ Chir~   4.54  2.04   1.21       2    45    22.3       1578 1.31e7
4 Italy  Buga~ Cent~   4.54  2.05   1.2        2     0    22.5       1600 3.30e7
5 UK     Bent~ Flyi~   5.32  2.01   1.48       5   351    10.9        521 1.3 e6
6 UK     Bent~ Cont~   4.81  2.23   1.40       5   235    10.9        521 1.15e6
7 UK     McLa~ Senna   4.74  2.15   1.20       2     0    12.4        800 5   e6
8 UK     McLa~ 720S    4.54  2.06   1.20       2     0    10.7        710 1.12e6
# i 5 more variables: mass <dbl>, performance <dbl>, type <fct>,
#   enginetype <fct>, origin2 <chr>
```

```r
# removing outliers and fitting the model again
fit3 <- lm(economy ~ horsepower + mass + origin2 + enginetype, data = data[-outs, ])
summary(fit3)
```

```
Call:
lm(formula = economy ~ horsepower + mass + origin2 + enginetype,
    data = data[-outs, ])

Residuals:
    Min      1Q  Median      3Q     Max
-5.5114 -1.0412 -0.0239  0.7882  8.3755

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)         -1.3332858  1.1048540  -1.207  0.23080
horsepower           0.0085139  0.0016137   5.276 9.55e-07 ***
mass                 0.0031681  0.0004994   6.344 9.74e-09 ***
origin2Non-European  1.0142196  0.4650532   2.181  0.03189 *
```

```
enginetypePetrol     1.6386479  0.5698544   2.876  0.00507 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.84 on 87 degrees of freedom
Multiple R-squared:  0.6146,    Adjusted R-squared:  0.5969
F-statistic: 34.68 on 4 and 87 DF,  p-value: < 2.2e-16
```

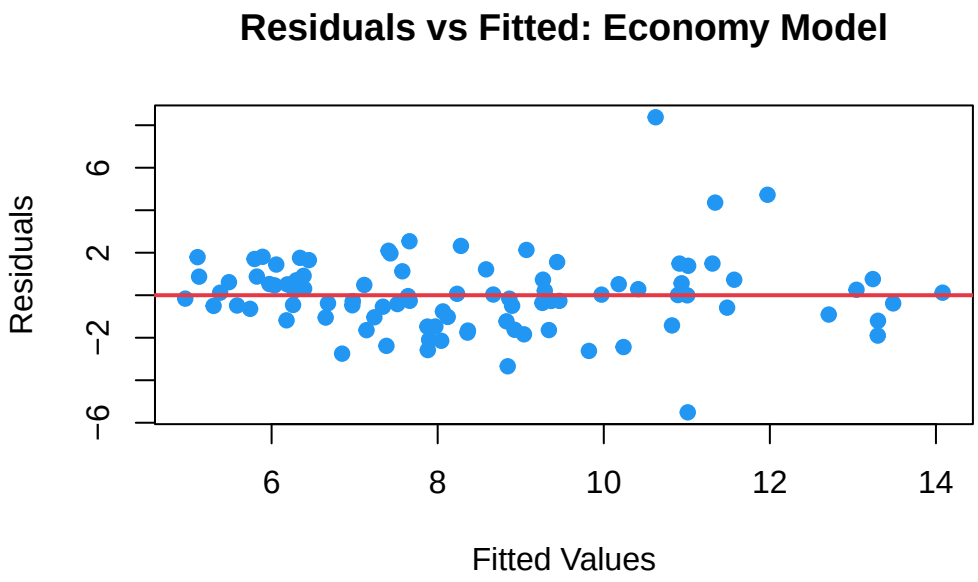
```r
qqnorm(fit3$residuals, main = "Q-Q Plot: Economy Model")
qqline(fit3$residuals, col = colors[1])
```




```r
shapiro.test(fit3$residuals)
```

```
    Shapiro-Wilk normality test

data:  fit3$residuals
W = 0.92514, p-value = 5.468e-05
```

```r
# non constant variance test
plot(fit3$fitted.values, fit3$residuals,
     main = "Residuals vs Fitted: Economy Model",
     xlab = "Fitted Values", ylab = "Residuals",
     col = colors[2], pch = 19)
abline(h = 0, col = colors[1], lwd = 2)
```




```r
# trying log transformation for economy
fit4 <- lm(log(economy) ~ length + width + height + seating + trunk +
    horsepower + price + mass + performance + origin2 +
    type + enginetype, data = data[-outs, ])
```

```r
fit5 <- step(fit4, direction = "both")
```

```r
summary(fit5)
```

```
Call:
lm(formula = log(economy) ~ width + trunk + horsepower + mass +
    origin2 + enginetype, data = data[-outs, ])

Residuals:
     Min       1Q   Median       3Q      Max
-0.53369 -0.13262  0.00556  0.13255  0.63377

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)          3.159e-01  4.236e-01   0.746  0.45788
width                4.518e-01  2.452e-01   1.843  0.06884 .
trunk               -1.552e-04  8.642e-05  -1.795  0.07615 .
horsepower           7.975e-04  1.965e-04   4.058  0.00011 ***
mass                 3.133e-04  5.884e-05   5.324 8.16e-07 ***
origin2Non-European  1.082e-01  5.050e-02   2.143  0.03499 *
```

```
enginetypePetrol     1.939e-01  6.161e-02   3.148  0.00227 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1976 on 85 degrees of freedom
Multiple R-squared:  0.6421,    Adjusted R-squared:  0.6168
F-statistic: 25.42 on 6 and 85 DF,  p-value: < 2.2e-16
```

```r
# just horsepower, mass and engine type
fit6 <- lm(log(economy) ~ horsepower + mass + enginetype , data = data[-outs, ])
summary(fit6)
```

```
Call:
lm(formula = log(economy) ~ horsepower + mass + enginetype, data = data[-outs,
    ])

Residuals:
     Min       1Q   Median       3Q      Max
-0.60402 -0.13313 -0.01347  0.12192  0.57666

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.141e+00  1.099e-01  10.389  < 2e-16 ***
horsepower       8.739e-04  1.655e-04   5.282 9.14e-07 ***
mass             3.153e-04  5.532e-05   5.699 1.58e-07 ***
enginetypePetrol 1.928e-01  6.309e-02   3.056  0.00297 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2046 on 88 degrees of freedom
Multiple R-squared:  0.6027,    Adjusted R-squared:  0.5891
F-statistic:  44.5 on 3 and 88 DF,  p-value: < 2.2e-16
```

```r
summary(aov(fit6))
```

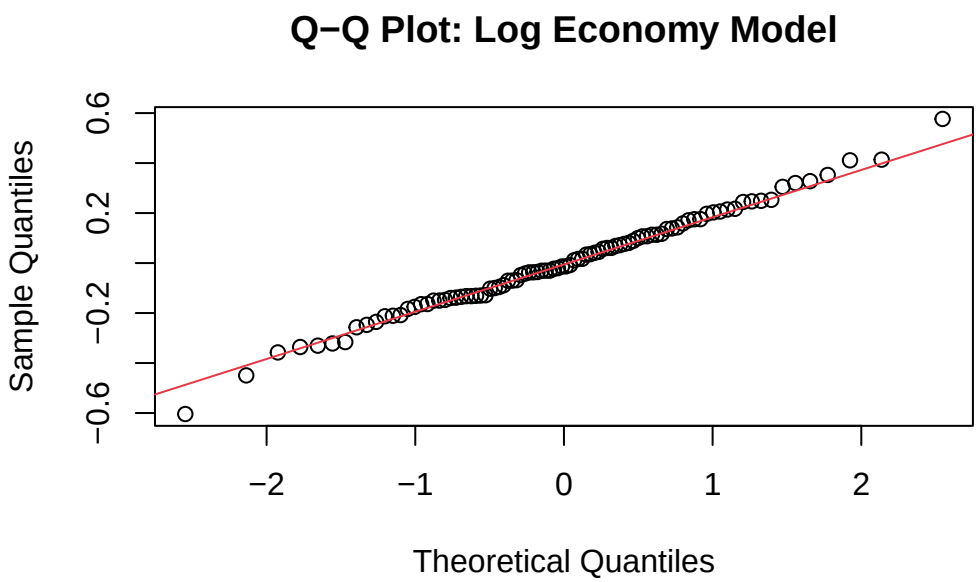
```
            Df Sum Sq Mean Sq F value   Pr(>F)
horsepower   1  4.084   4.084  97.561 6.42e-16 ***
mass         1  1.113   1.113  26.586 1.54e-06 ***
enginetype   1  0.391   0.391   9.342  0.00297 **
Residuals   88  3.684   0.042
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
shapiro.test(fit6$residuals)
```

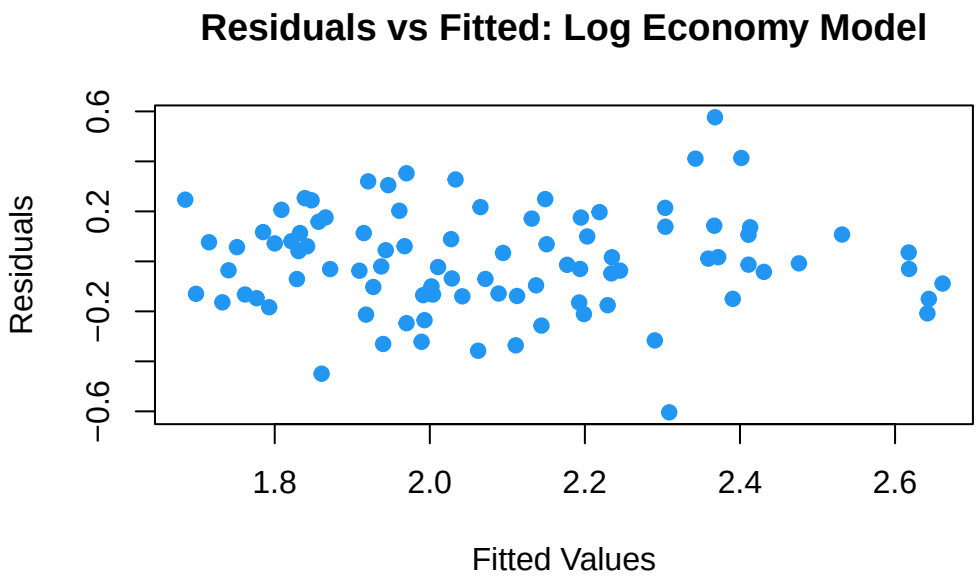
```
    Shapiro-Wilk normality test
```

```
data:  fit6$residuals
W = 0.99472, p-value = 0.9766
```

```r
qqnorm(fit6$residuals, main = "Q-Q Plot: Log Economy Model")
qqline(fit6$residuals, col = colors[1])
```




```r
plot(fit6$fitted.values, fit6$residuals,
     main = "Residuals vs Fitted: Log Economy Model",
     xlab = "Fitted Values", ylab = "Residuals",
     col = colors[2], pch = 19)
```



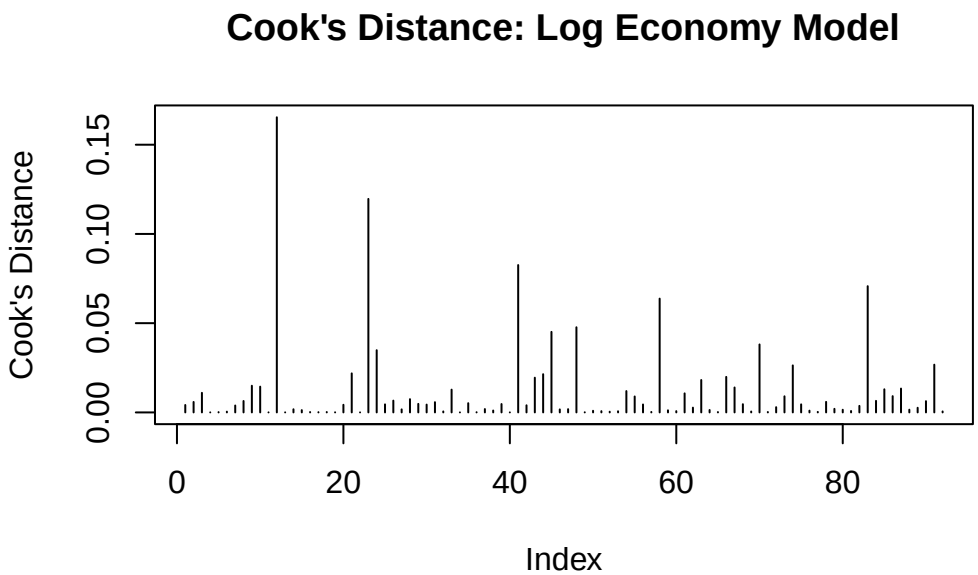

```r
vif(fit6)
```

```
horsepower       mass enginetype
  1.440121   1.510386   1.061534
```

```r
outlierTest(fit6)
```

```
No Studentized residuals with Bonferroni p < 0.05
Largest |rstudent|:
    rstudent unadjusted p-value Bonferroni p
48 -3.129234          0.0023854      0.21946
```

```r
cooksd <- cooks.distance(fit6)
plot(cooksd, type = "h", main = "Cook's Distance: Log Economy Model", ylab = "Cook's Distance")
```




```r
which(cooksd > 4/nrow(data))
```

```
12 23 41 45 48 58 83
12 23 41 45 48 58 83
```

```r
# 12 23 41 48 61 74 83
# removing outliers and fitting the model again
fit7 <- lm(log(economy) ~ horsepower + mass + enginetype , data = data[-c(outs, 12, 23, 41, 48, 61, 7
summary(fit7)
```

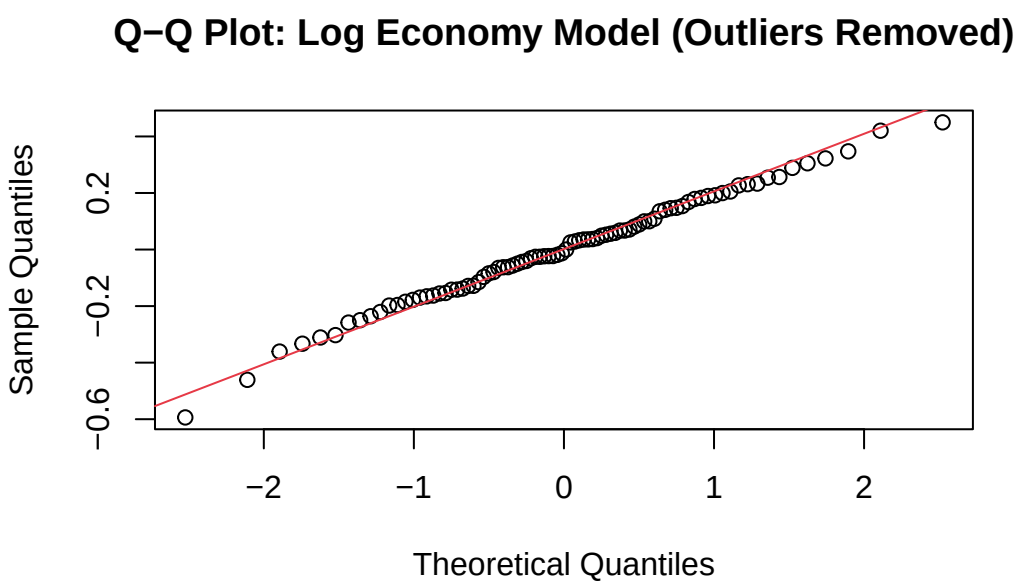
```
Call:
lm(formula = log(economy) ~ horsepower + mass + enginetype, data = data[-c(outs,
    12, 23, 41, 48, 61, 74, 83), ])

Residuals:
     Min       1Q   Median       3Q      Max
-0.59379 -0.13568 -0.00662  0.13925  0.44989

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.214e+00  1.087e-01  11.170  < 2e-16 ***
horsepower       9.028e-04  1.626e-04   5.551 3.41e-07 ***
mass             2.749e-04  5.537e-05   4.965 3.68e-06 ***
enginetypePetrol 1.813e-01  6.098e-02   2.973  0.00387 **
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1963 on 82 degrees of freedom
Multiple R-squared:  0.6016,    Adjusted R-squared:  0.587
F-statistic: 41.27 on 3 and 82 DF,  p-value: 2.349e-16
```

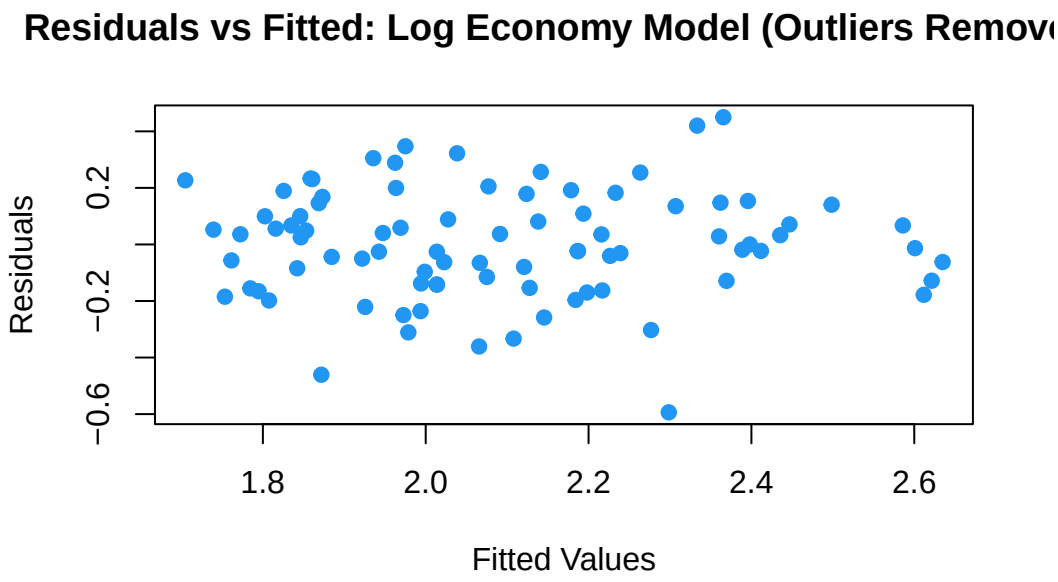
```r
shapiro.test(fit7$residuals)
```

```
    Shapiro-Wilk normality test

data:  fit7$residuals
W = 0.99319, p-value = 0.9383
```

```r
qqnorm(fit7$residuals, main = "Q-Q Plot: Log Economy Model (Outliers Removed)")
qqline(fit7$residuals, col = colors[1])
```




```r
# normally distributed residuals observed

plot(fit7$fitted.values, fit7$residuals,
     main = "Residuals vs Fitted: Log Economy Model (Outliers Removed)",
     xlab = "Fitted Values", ylab = "Residuals",
     col = colors[2], pch = 19)
```



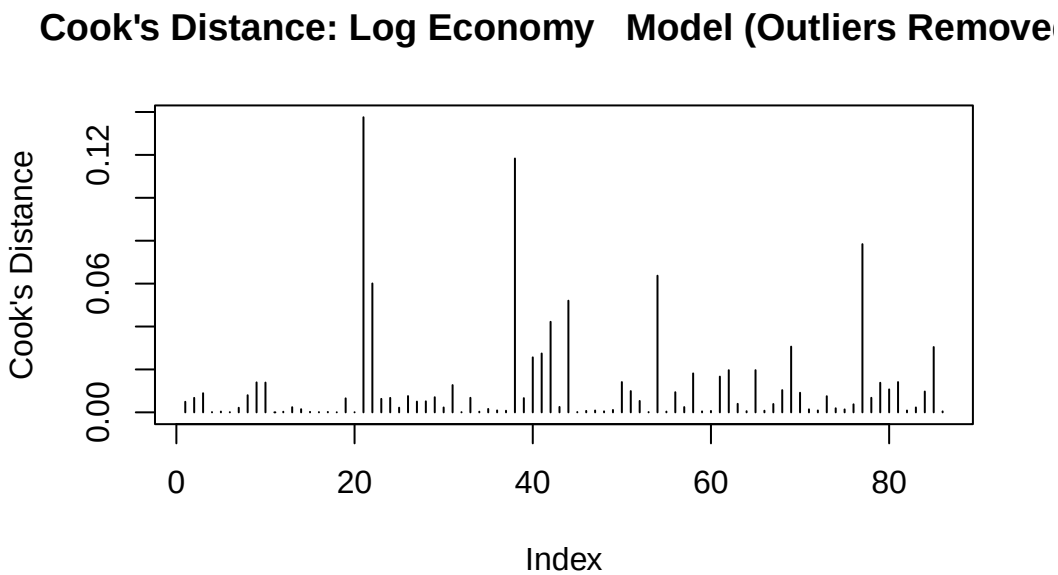

```r
# no funnel shape observed, variance is somewhat constant now

vif(fit7)
```

```
horsepower       mass enginetype
  1.443204   1.520693   1.065232
```

```r
# all variables are significant and no multicollinearity issue (VIF < 5)

outlierTest(fit7)
```

```
No Studentized residuals with Bonferroni p < 0.05
Largest |rstudent|:
    rstudent unadjusted p-value Bonferroni p
44 -3.229874          0.0017902      0.15395
```

```r
cooksd <- cooks.distance(fit7)
plot(cooksd, type = "h",
main = "Cook's Distance: Log Economy   Model (Outliers Removed)",
ylab = "Cook's Distance")
```




```r
which(cooksd > 4/nrow(data))
```

```
21 22 38 42 44 54 77
21 22 38 42 44 54 77
```

```r
# comparing fit6 and fit7, very slight improvement in R^2 and adjusted R^2
# going with fit6 for economy model
```

Summary of Economy Analysis

- The final model for economy includes `horsepower`, `mass`, and `enginetype` as significant predictors.
- The model explains a reasonable amount of variance in fuel economy, and the residuals are approximately normally distributed with no significant heteroscedasticity observed after removing outliers.
- The model can be represented as:

  $$\log(economy) = 1.141 + 0.0008739 \cdot horsepower + 0.0003153 \cdot mass + 0.1928 \cdot enginetypePetrol$$

5.2 Visualizations

```
`geom_smooth()` using formula = 'y ~ x'
```

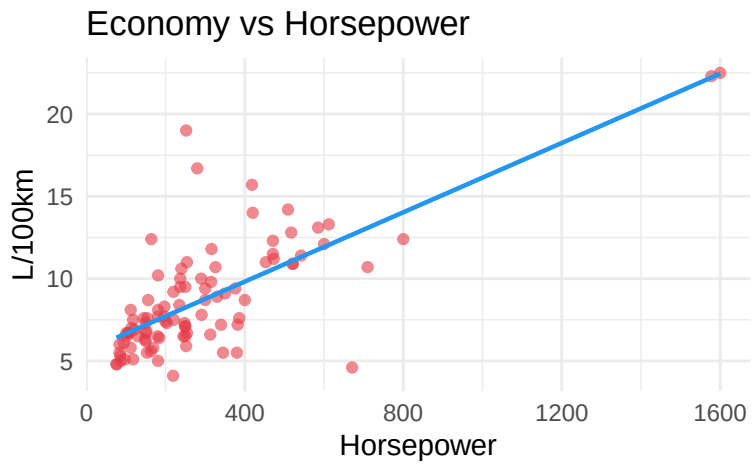


Figure 2: Relationships Between Economy and Key Variables

```
`geom_smooth()` using formula = 'y ~ x'
```

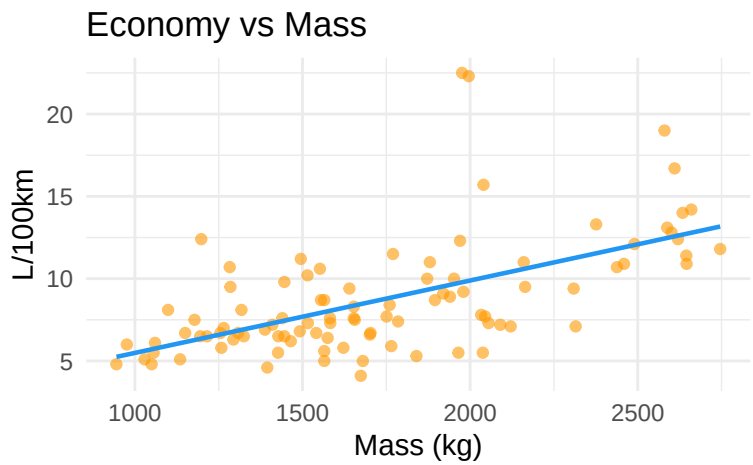


Figure 3: Relationships Between Economy and Key Variables

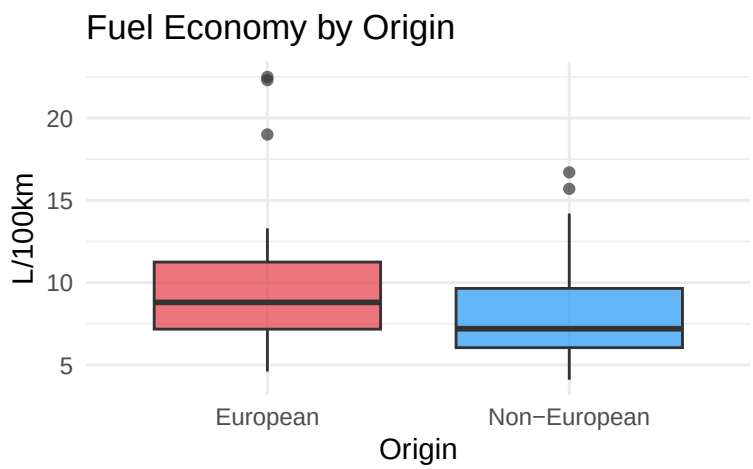


Figure 4: Relationships Between Economy and Key Variables

```
`geom_smooth()` using formula = 'y ~ x'
```

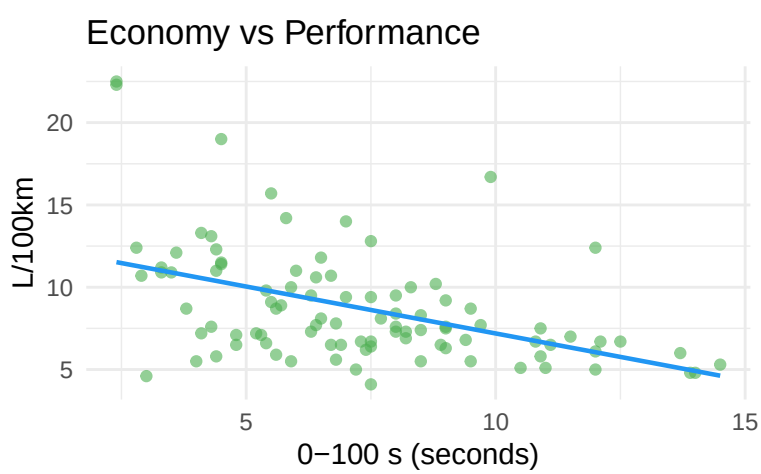


Figure 5: Relationships Between Economy and Key Variables

6 Price Analysis

```
# price histogram
ggplot(data, aes(x = price)) +
  geom_histogram(fill = colors[3], bins = 30) +
  labs(title = "Distribution of Car Prices", x = "Price (QAR)", y = "Frequency") +
  theme_minimal()
```

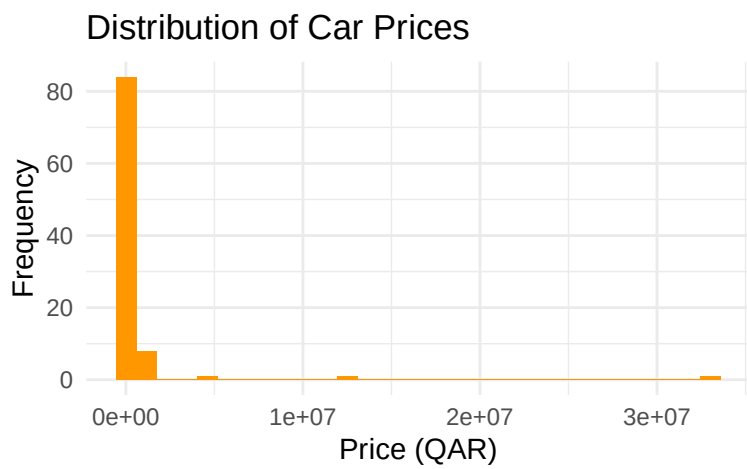


Figure 6: Price Distribution Analysis

```
# log price histogram
ggplot(data, aes(x = log(price))) +
  geom_histogram(fill = colors[3], bins = 30) +
  labs(title = "Distribution of Log Car Prices", x = "Log Price", y = "Frequency") +
  theme_minimal()
```

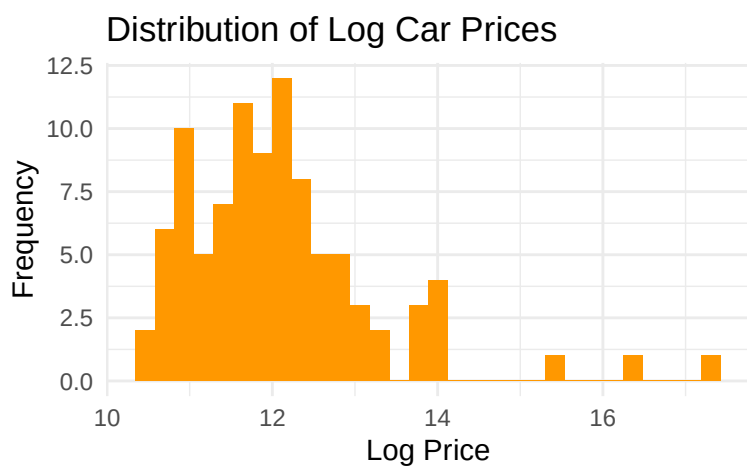


Figure 7: Price Distribution Analysis

```
# removing outliers for price (any price above 600000)
sub_data <- data |> filter(price < 600000)
ggplot(sub_data, aes(x = price)) +
  geom_histogram(fill = colors[4], bins = 30) +
  labs(title = "Distribution of Car Prices (Extreme values Removed)",
```

```
  x = "Price (QAR)", y = "Frequency") +
  theme_minimal()
```

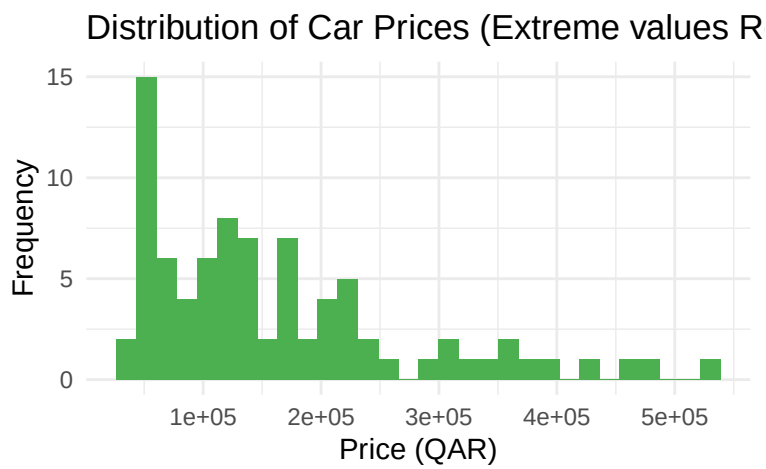


Figure 8: Price Distribution Analysis

```
# plotting all three histograms together, more height
```

6.1 Model Development

```
fit1 <- lm(log(price) ~ length + width + height + seating +
    trunk + horsepower + economy + mass + performance +
    origin2 + type + enginetype, data = data[-outs, ])
summary(fit1)
```

```
Call:
lm(formula = log(price) ~ length + width + height + seating +
    trunk + horsepower + economy + mass + performance + origin2 +
    type + enginetype, data = data[-outs, ])

Residuals:
     Min       1Q   Median       3Q      Max
-0.86372 -0.12689  0.04941  0.14885  0.60996

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)          9.889e+00  9.863e-01  10.027 1.29e-15 ***
length               2.025e-02  1.771e-01   0.114  0.90926
width                1.054e+00  4.229e-01   2.494  0.01479 *
```

```
height              -9.169e-01  4.565e-01  -2.008  0.04810 *
seating              1.863e-02  4.842e-02   0.385  0.70146
trunk               -4.941e-05  1.468e-04  -0.336  0.73743
horsepower           4.296e-03  4.239e-04  10.134 8.02e-16 ***
economy             -1.764e-02  1.740e-02  -1.014  0.31395
mass                 2.766e-04  1.460e-04   1.894  0.06192 .
performance         -9.399e-04  2.012e-02  -0.047  0.96287
origin2Non-European -3.065e-01  8.587e-02  -3.569  0.00062 ***
typeHatchback       -8.007e-02  1.904e-01  -0.420  0.67533
typeSUV              1.543e-01  2.008e-01   0.768  0.44455
typeSedan           -2.049e-02  1.544e-01  -0.133  0.89477
enginetypePetrol     7.118e-02  9.691e-02   0.735  0.46485
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2833 on 77 degrees of freedom
Multiple R-squared:  0.9216,    Adjusted R-squared:  0.9074
F-statistic:  64.7 on 14 and 77 DF,  p-value: < 2.2e-16
```

```r
# stepwise selection
fit2 <- step(fit1, direction = "both")
```

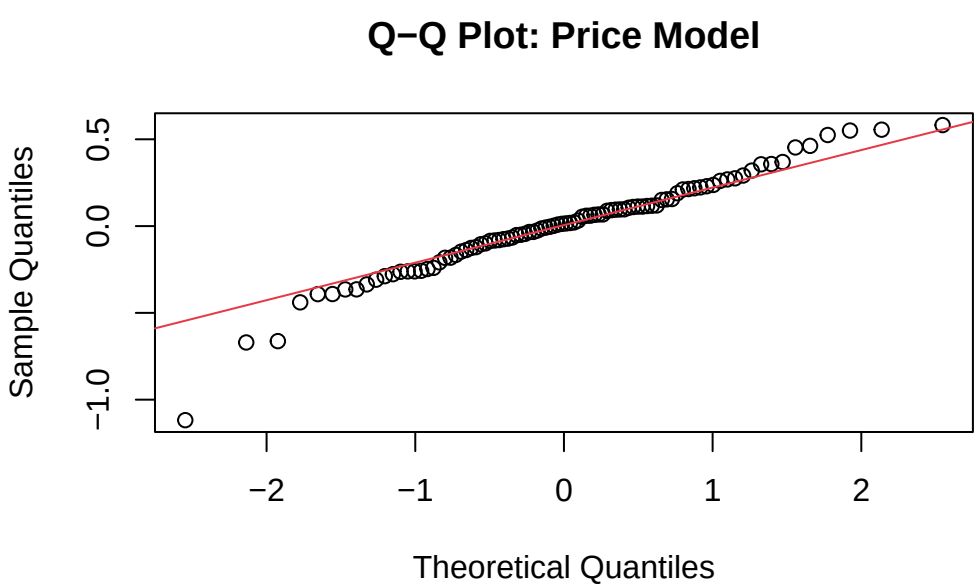
```r
summary(fit2)
```

```
Call:
lm(formula = log(price) ~ width + height + horsepower + mass +
    origin2, data = data[-outs, ])

Residuals:
     Min       1Q   Median       3Q      Max
-1.02760 -0.12519  0.02291  0.15361  0.63518

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)          9.2318910  0.6146638  15.019  < 2e-16 ***
width                1.2144622  0.3395876   3.576 0.000575 ***
height              -0.4729159  0.2324164  -2.035 0.044953 *
horsepower           0.0040774  0.0002880  14.159  < 2e-16 ***
mass                 0.0002057  0.0001137   1.808 0.074092 .
origin2Non-European -0.3489595  0.0718550  -4.856  5.3e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2771 on 86 degrees of freedom
Multiple R-squared:  0.9163,    Adjusted R-squared:  0.9114
F-statistic: 188.2 on 5 and 86 DF,  p-value: < 2.2e-16
```

```r
fit3 <- lm(log(price) ~ width + horsepower + origin2, data = data[-outs, ])
summary(fit3)
```

```
Call:
lm(formula = log(price) ~ width + horsepower + origin2, data = data[-outs,
    ])

Residuals:
     Min       1Q   Median       3Q      Max
-1.11823 -0.14034  0.01406  0.15137  0.58199

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)          8.7117554  0.5669747  15.365  < 2e-16 ***
width                1.2615622  0.3180916   3.966 0.000149 ***
horsepower           0.0043083  0.0002676  16.098  < 2e-16 ***
origin2Non-European -0.3864812  0.0705142  -5.481 3.98e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2808 on 88 degrees of freedom
Multiple R-squared:  0.912, Adjusted R-squared:  0.909
F-statistic:   304 on 3 and 88 DF,  p-value: < 2.2e-16
```

```r
qqnorm(fit3$residuals, main = "Q-Q Plot: Price Model")
qqline(fit3$residuals, col = colors[1])
```



```r
shapiro.test(fit3$residuals)
```

```
    Shapiro-Wilk normality test

data:  fit3$residuals
W = 0.96105, p-value = 0.007626
```

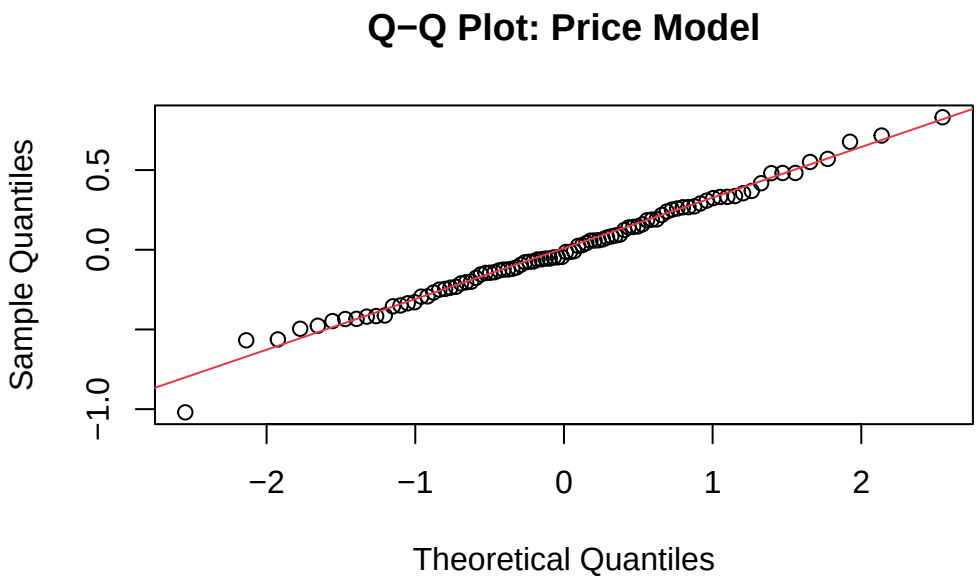
```r
# removing origin2 variable

fit4 <- lm(log(price) ~ horsepower + width, data = data[-outs, ])
summary(fit4)
```

```
Call:
lm(formula = log(price) ~ horsepower + width, data = data[-outs,
    ])

Residuals:
     Min       1Q   Median       3Q      Max
-1.02005 -0.20542 -0.02927  0.22351  0.83143

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 8.0366048  0.6373588  12.609  < 2e-16 ***
horsepower  0.0048199  0.0002889  16.686  < 2e-16 ***
width       1.4170069  0.3648709   3.884 0.000197 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3234 on 89 degrees of freedom
Multiple R-squared:  0.882, Adjusted R-squared:  0.8793
F-statistic: 332.5 on 2 and 89 DF,  p-value: < 2.2e-16
```

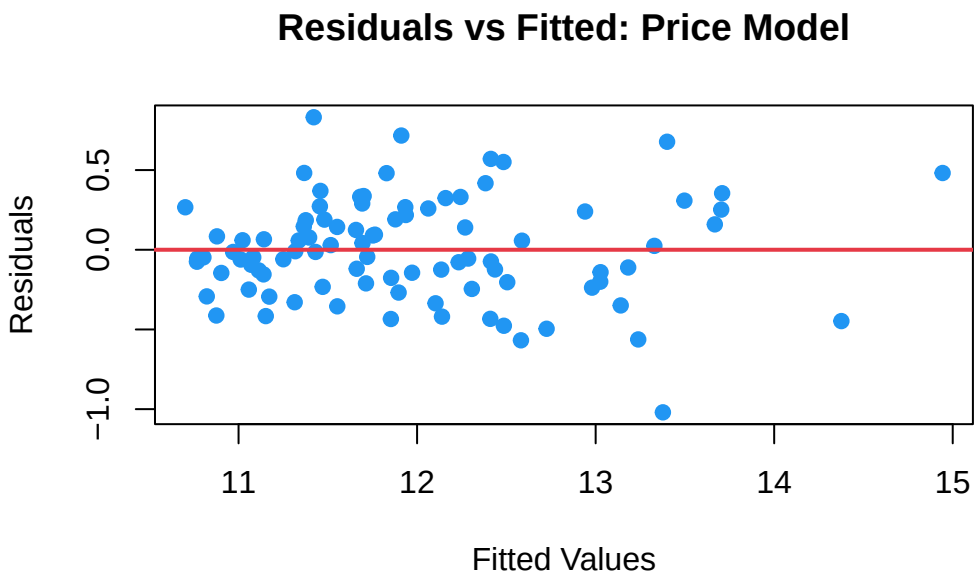
```r
shapiro.test(fit4$residuals)
```

```
    Shapiro-Wilk normality test

data:  fit4$residuals
W = 0.99169, p-value = 0.8377
```

```r
qqnorm(fit4$residuals, main = "Q-Q Plot: Price Model")
qqline(fit4$residuals, col = colors[1])
```



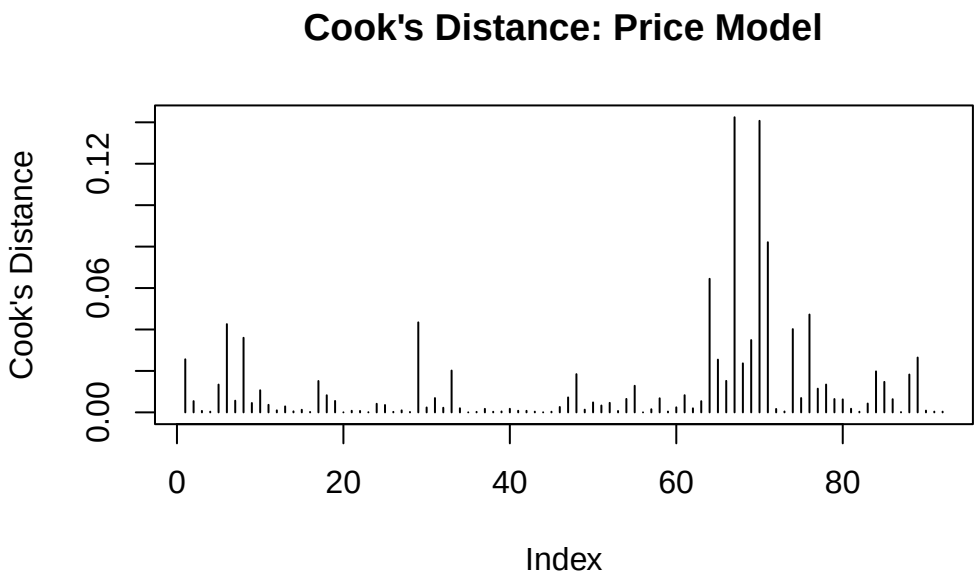

```r
residuals <- fit4$residuals
plot(fit4$fitted.values, residuals,
     main = "Residuals vs Fitted: Price Model",
     xlab = "Fitted Values", ylab = "Residuals",
     col = colors[2], pch = 19)
abline(h = 0, col = colors[1], lwd = 2)
```




```r
vif(fit4)
```

```
horsepower      width
   1.75659    1.75659
```

```r
cooksd <- cooks.distance(fit4)
plot(cooksd, type = "h", main = "Cook's Distance: Price Model", ylab = "Cook's Distance")
```




```r
which(cooksd > 4/nrow(data))
```

```
 6 29 64 67 70 71 76
 6 29 64 67 70 71 76
```

```r
# this will lead to endless loop of outliers, so we can stop here

# simple model with just horsepower
fit5 <- lm(log(price) ~ horsepower, data = data[-outs, ])
summary(fit5)
```

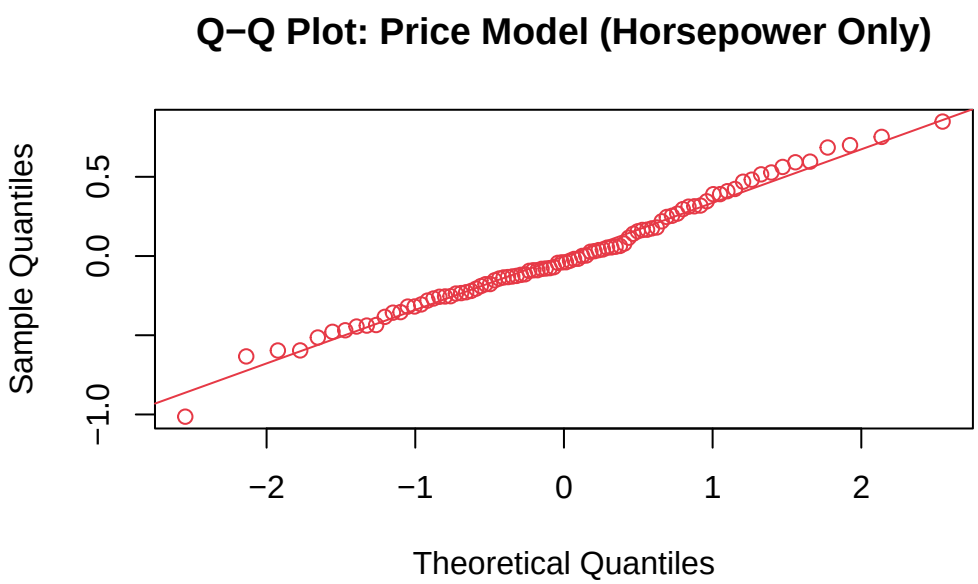
```
Call:
lm(formula = log(price) ~ horsepower, data = data[-outs, ])

Residuals:
     Min       1Q   Median       3Q      Max 
-1.01403 -0.22984 -0.03946  0.22590  0.84868 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 1.050e+01  7.201e-02  145.78   <2e-16 ***
horsepower  5.556e-03  2.344e-04   23.71   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.3478 on 90 degrees of freedom
Multiple R-squared:  0.862, Adjusted R-squared:  0.8604
F-statistic:   562 on 1 and 90 DF,  p-value: < 2.2e-16
```

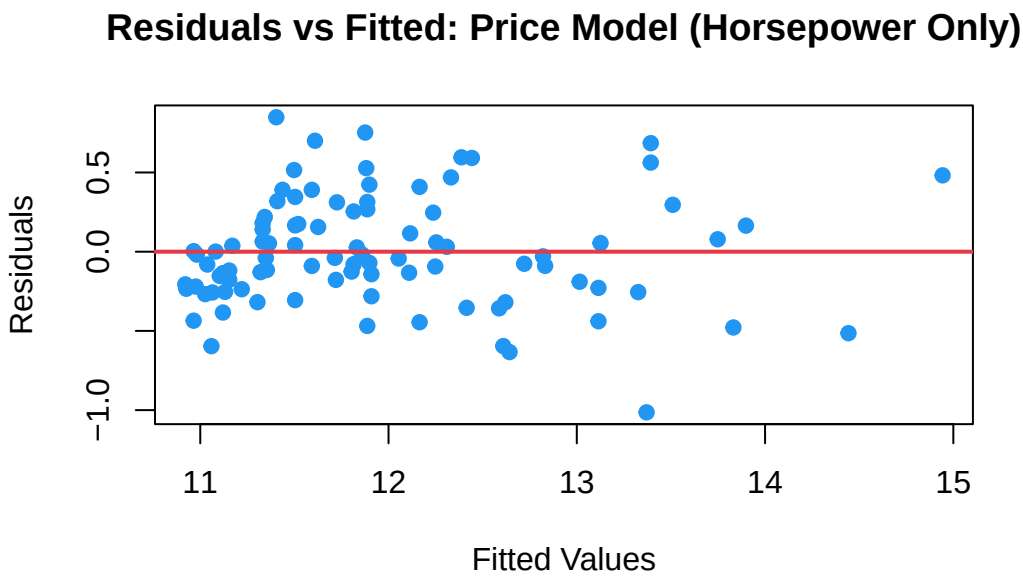
```r
shapiro.test(fit5$residuals)
```

```
    Shapiro-Wilk normality test

data:  fit5$residuals
W = 0.9879, p-value = 0.5621
```

```r
qqnorm(fit5$residuals, main = "Q-Q Plot: Price Model (Horsepower Only)", col = colors[1])
qqline(fit5$residuals, col = colors[1])
```




```r
residuals <- fit5$residuals
plot(fit5$fitted.values, residuals,
     main = "Residuals vs Fitted: Price Model (Horsepower Only)",
     xlab = "Fitted Values", ylab = "Residuals",
     col = colors[2], pch = 19)
abline(h = 0, col = colors[1], lwd = 2)
```




```
# slight funnel shape observed, but a sim
# fit5 is the final price model
```

Summary of Price Analysis

- The final model for price includes `horsepower` as the most significant predictor, with `width` and `origin2` being less significant. However, removing `origin2` did not significantly improve the model, so we can go with the simpler model with just `horsepower`.
- Some heteroscedasticity is observed in the residuals, but the model still provides a reasonable fit to the data.
- The model can be represented as:

$$\log(price) = 10.50 + 0.005556 \cdot horsepower$$

6.2 Visualizations

```
`geom_smooth()` using formula = 'y ~ x'
```

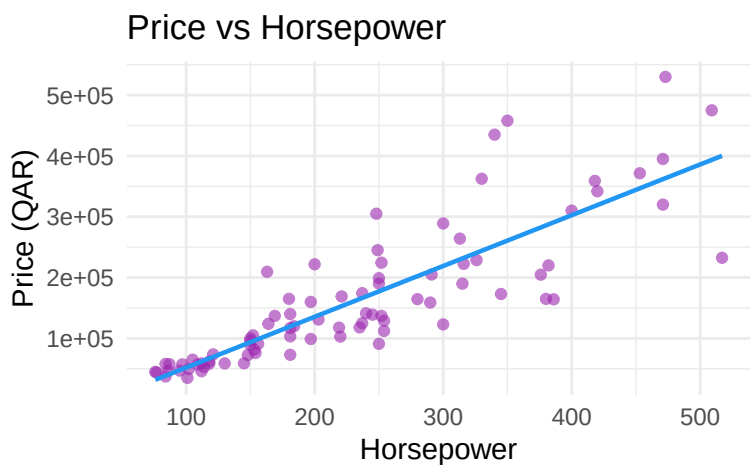


Figure 9: Relationships Between Price and Key Variables

```
`geom_smooth()` using formula = 'y ~ x'
```

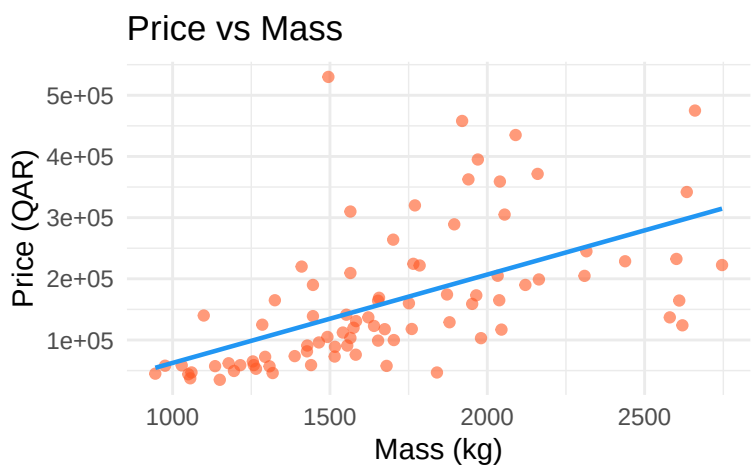


Figure 10: Relationships Between Price and Key Variables

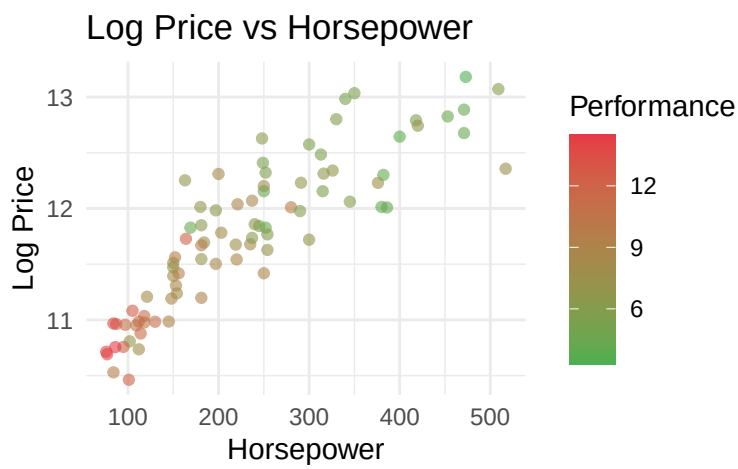


Figure 11: Relationships Between Price and Key Variables

```
`geom_smooth()` using formula = 'y ~ x'
```

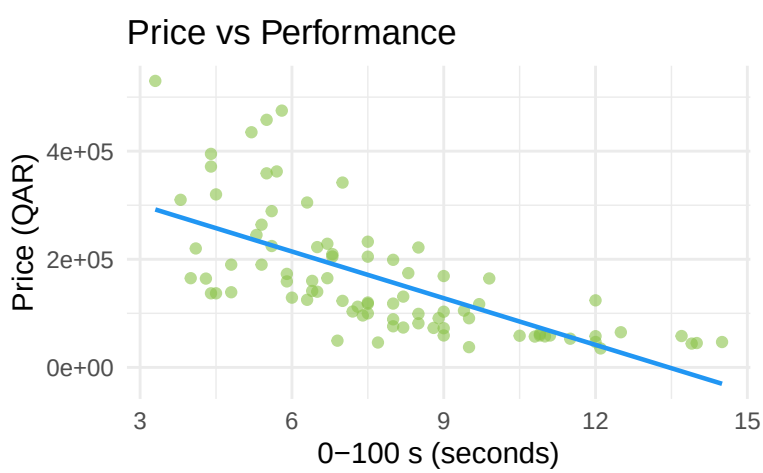


Figure 12: Relationships Between Price and Key Variables

7 Bayesian Approach

```
library(brms)
library(bayesplot)
```

7.1 Bayesian linear regression for price model

```r
final_fit_price <- fit5
summary(final_fit_price)
```

```
Call:
lm(formula = log(price) ~ horsepower, data = data[-outs, ])

Residuals:
     Min       1Q   Median       3Q      Max 
-1.01403 -0.22984 -0.03946  0.22590  0.84868 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 1.050e+01  7.201e-02  145.78   <2e-16 ***
horsepower  5.556e-03  2.344e-04   23.71   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3478 on 90 degrees of freedom
Multiple R-squared:  0.862, Adjusted R-squared:  0.8604 
F-statistic:   562 on 1 and 90 DF,  p-value: < 2.2e-16
```

```r
brm_price <- brm(log(price) ~ horsepower, data = data[-outs, ],
                 prior = c(set_prior("normal(0, 10)", class = "b"),
                           set_prior("normal(0, 10)", class = "Intercept"),
                           set_prior("exponential(1)", class = "sigma")),
                 iter = 4000, warmup = 1000, chains = 4, seed = 123,
                 silent = 1, refresh = 0)
```

```r
summary(brm_price)
```

```
 Family: gaussian
  Links: mu = identity
Formula: log(price) ~ horsepower 
   Data: data[-outs, ] (Number of observations: 92) 
  Draws: 4 chains, each with iter = 4000; warmup = 1000; thin = 1;
         total post-warmup draws = 12000

Regression Coefficients:
           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     10.50      0.07    10.35    10.65 1.00     8476     7980
horsepower     0.01      0.00     0.01     0.01 1.00    12943     8031

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
```

```
sigma     0.35      0.03     0.31     0.41 1.00     4575     4557

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
bayes_R2(brm_price)
```

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.8600546 0.01105098 0.8331697 0.8755178
```

```r
plot(brm_price)
```

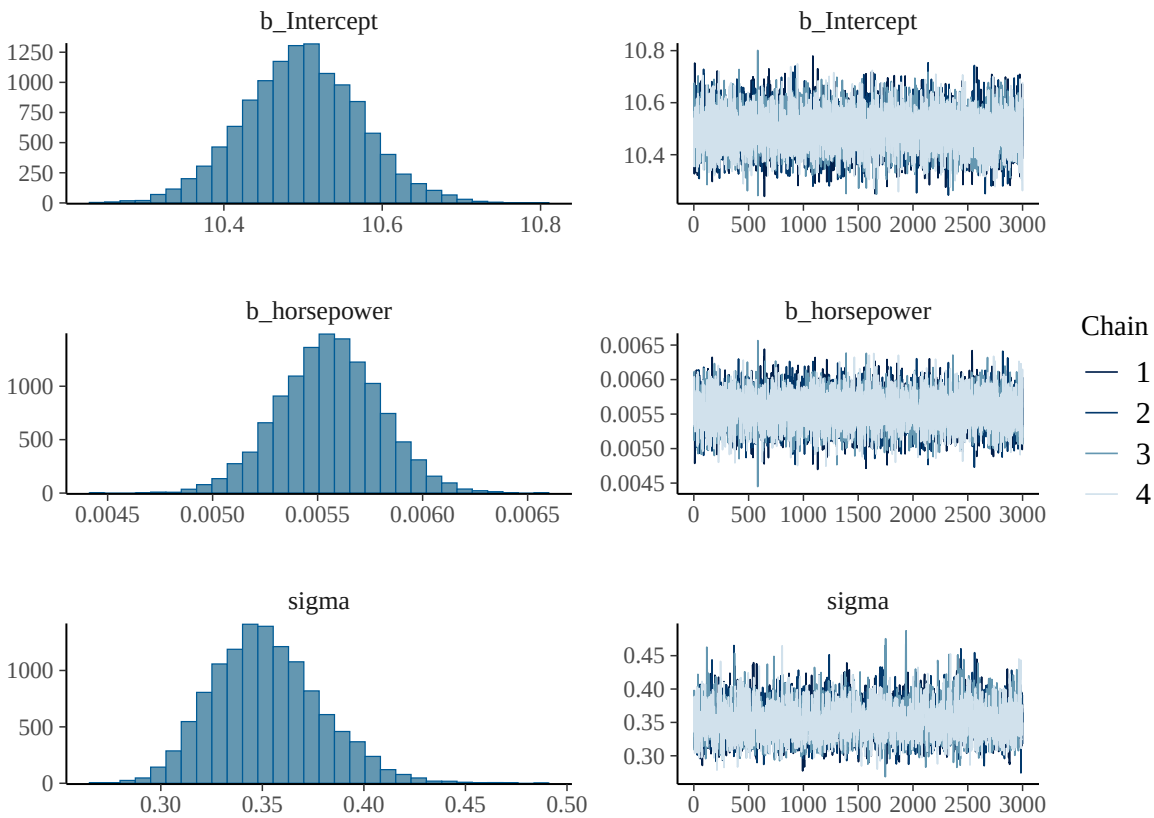


Figure 13: Bayesian Model Diagnostics for Price Model

```r
pp_check(brm_price)
```

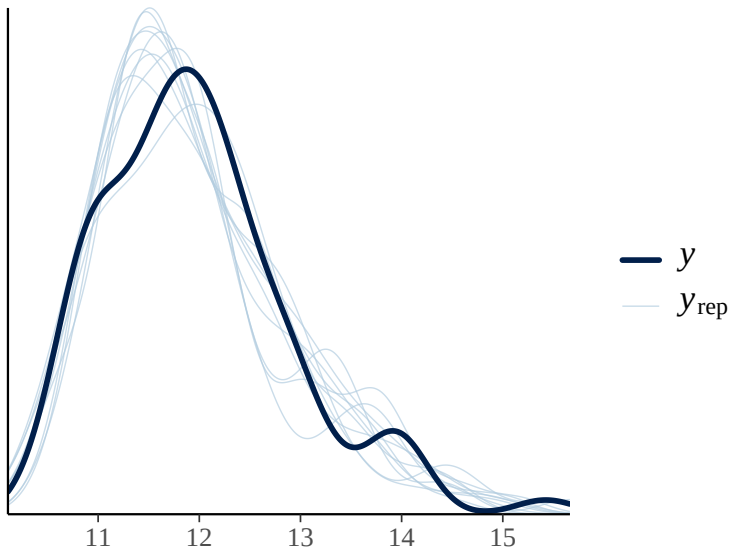
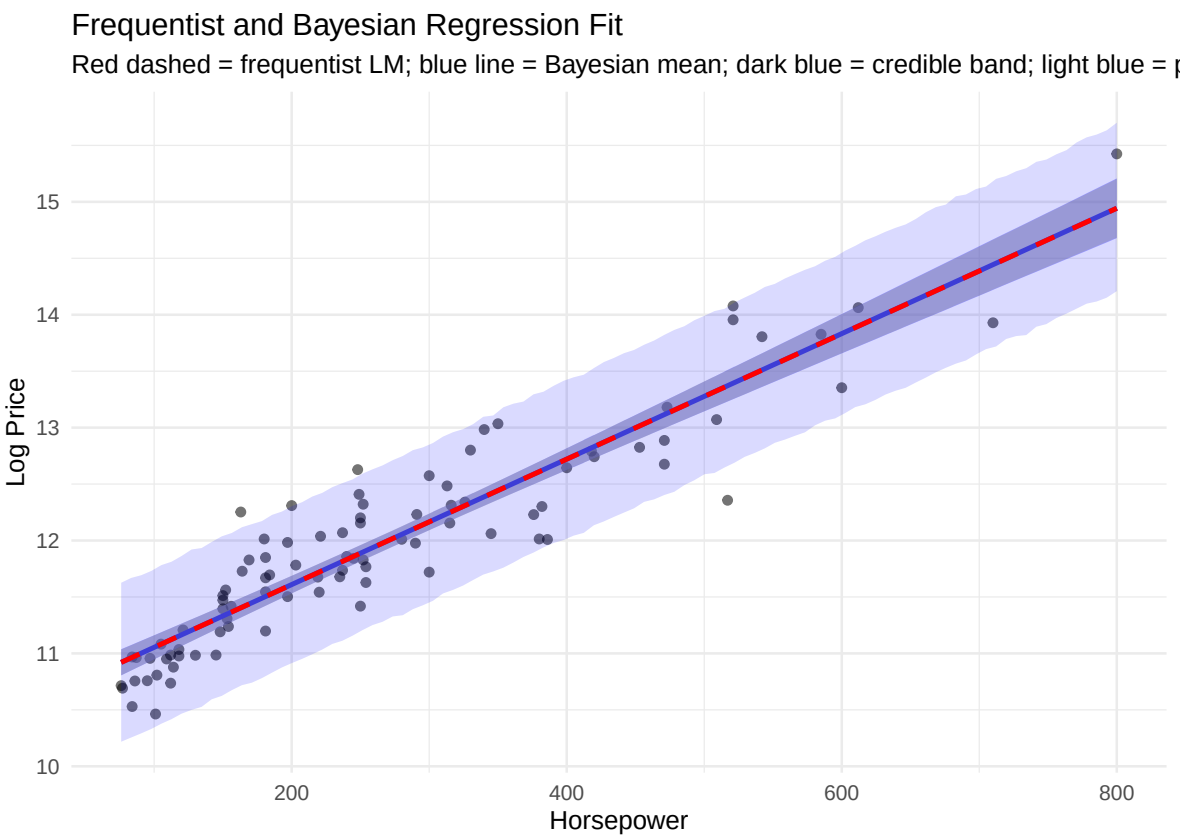


Figure 14: Posterior Predictive Check for Price Model



7.2 Bayesian linear regression for economy model

```r
final_fit_economy <- fit6
summary(final_fit_economy)
```

```
Call:
lm(formula = log(economy) ~ horsepower + mass + enginetype, data = data[-outs,
    ])

Residuals:
     Min       1Q   Median       3Q      Max 
-0.60402 -0.13313 -0.01347  0.12192  0.57666 

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)      1.141e+00  1.099e-01  10.389  < 2e-16 ***
horsepower       8.739e-04  1.655e-04   5.282 9.14e-07 ***
mass             3.153e-04  5.532e-05   5.699 1.58e-07 ***
enginetypePetrol 1.928e-01  6.309e-02   3.056  0.00297 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2046 on 88 degrees of freedom
Multiple R-squared:  0.6027,	Adjusted R-squared:  0.5891 
F-statistic:  44.5 on 3 and 88 DF,  p-value: < 2.2e-16
```

```r
brm_economy <- brm(log(economy) ~ horsepower + mass + enginetype,
                   data = data[-c(outs), ],
                   prior = c(set_prior("normal(0, 10)", class = "b"),
                             set_prior("normal(0, 10)", class = "Intercept"),
                             set_prior("exponential(1)", class = "sigma")),
                   iter = 4000, warmup = 1000, chains = 4, seed = 123,
                   silent = 1, refresh = 0)
```

```r
summary(brm_economy)
```

```
 Family: gaussian
  Links: mu = identity
Formula: log(economy) ~ horsepower + mass + enginetype
   Data: data[-c(outs), ] (Number of observations: 92)
  Draws: 4 chains, each with iter = 4000; warmup = 1000; thin = 1;
         total post-warmup draws = 12000

Regression Coefficients:
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept            1.14      0.11     0.92     1.36 1.00     7881     8237
```

```
horsepower           0.00      0.00     0.00     0.00 1.00    11970     7532
mass                 0.00      0.00     0.00     0.00 1.00    11938     9222
enginetypePetrol     0.19      0.06     0.07     0.32 1.00     5574     5026

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.21      0.02     0.18     0.24 1.00     5815     5301

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
bayes_R2(brm_economy)
```

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.5955237 0.04256884 0.4994463 0.6646089
```

```r
plot(brm_economy)
```

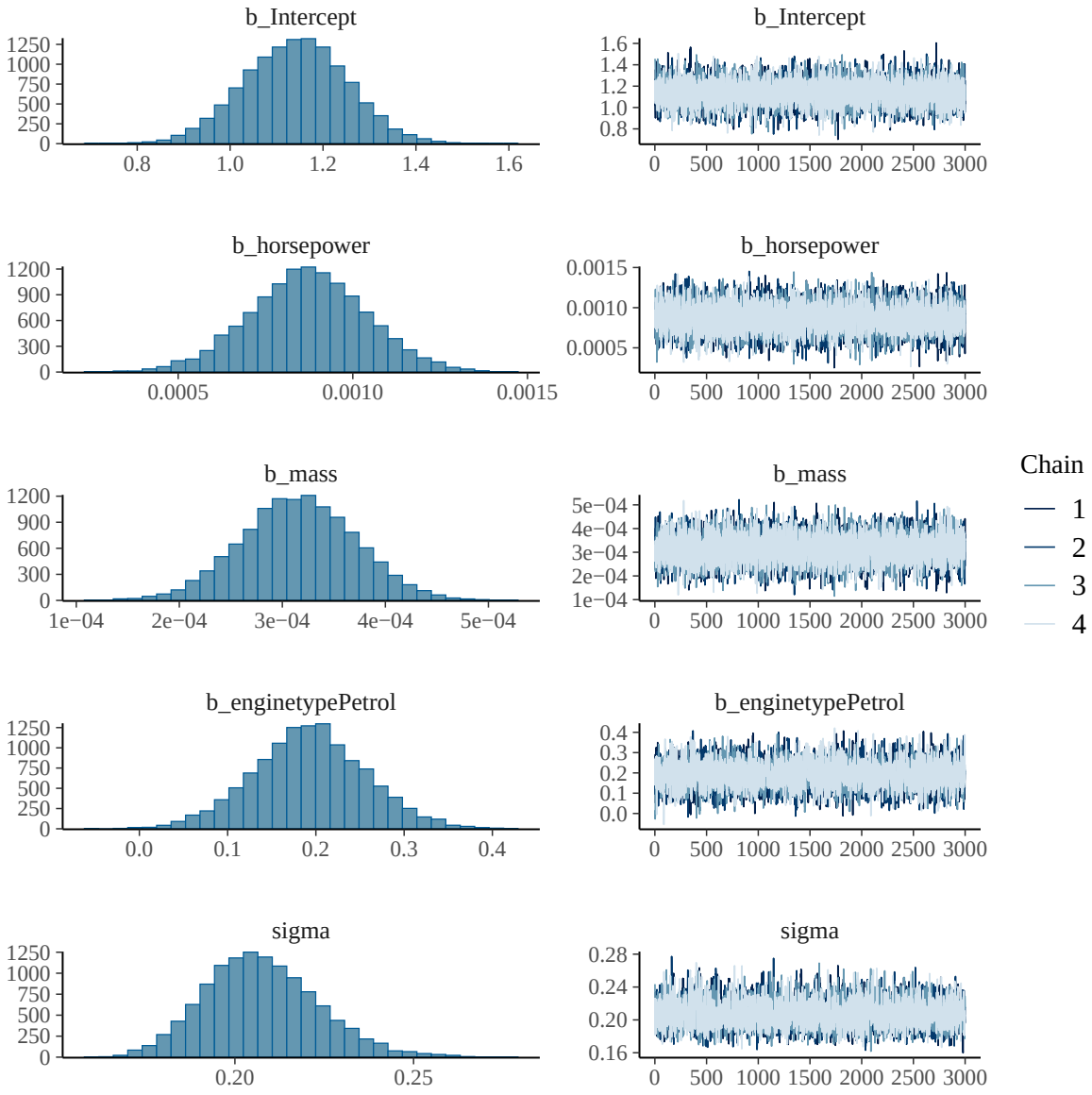


Figure 15: Bayesian Model Diagnostics for Economy Model

```
pp_check(brm_economy)
```

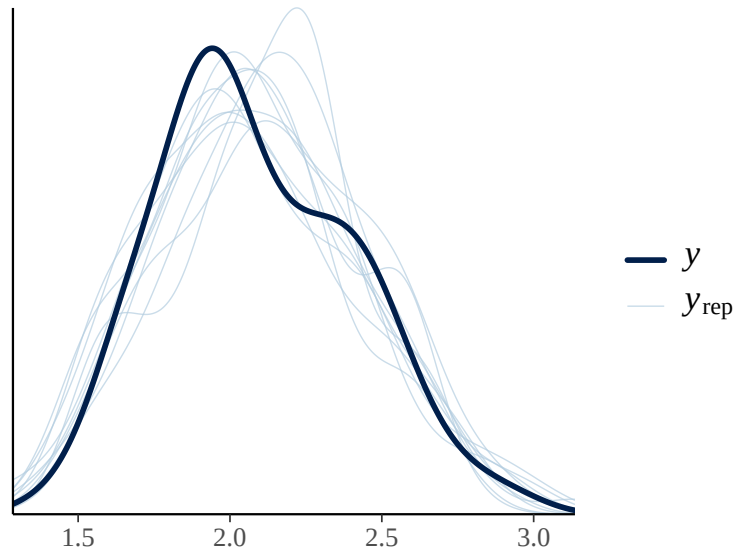


Figure 16: Posterior Predictive Check for Economy Model

8 Conclusions

The analysis of the `qatarcars` dataset revealed key insights into the determinants of fuel economy and price for cars in Qatar. For fuel economy, horsepower, mass, and engine type were significant predictors, with the final model explaining a reasonable amount of variance. For price, horsepower emerged as the most significant predictor, although some heteroscedasticity was observed in the residuals. The visualizations supported these findings, showing clear relationships between the key variables and the outcomes. Final model with $R^2 = 0.602$ (can explain 60.2% of the variance) can be represented as:

$$\log(economy) = 1.141 + 0.0008739 \cdot horsepower + 0.0003153 \cdot mass + 0.1928 \cdot enginetypePetrol$$

For price, horsepower emerged as the most significant predictor, although some heteroscedasticity was observed in the residuals. The visualizations supported these findings, showing clear relationships between the key variables and the outcomes. Final model with $R^2 = 0.862$ (can explain 86.2% of the variance) can be represented as:

$$\log(price) = 10.50 + 0.005556 \cdot horsepower$$

Overall, this analysis provides a comprehensive understanding of the determinants of car fuel economy and price in the Qatari market.

References

Musgrave, Paul. 2025. "Defaulting to Inclusion: Producing Sample Datasets for the Global Data Science Classroom." *Journal of Political Science Education*, October, 1–11. https://doi.org/10.1080/15512169.2025.2572320.